

From Retrieval to Reasoning

An Agentic Knowledge Graph Approach to Supply Chain Vulnerability

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Abstract – Global supply chains are opaque beyond tier-1: the multi-tier dependencies that harbour the most consequential disruptions remain invisible to the organisations that depend on them. We present an *Agentic Knowledge Graph* framework that transforms supply chain risk analysis from passive retrieval into active, multi-hop reasoning. A Neo4j knowledge graph, automatically populated via GLiNER named entity recognition and LLM-driven relation extraction from a corpus of public reports spanning corporate disclosures, commodity intelligence, and regulatory filings, is coupled with a LangGraph ReAct agent that fuses structural graph intelligence with live geopolitical signals from the Tavily search API. The agent traverses multi-tier supplier paths, computes a formula-based Criticality Index, and produces fully provenance-traced risk reports. Validated on Tesla’s cobalt supply chain as a representative case of critical-mineral EV battery risk, the system autonomously uncovers a three-hop DRC dependency path invisible to tier-1 analysis, cross-references a live cobalt export tax event, and proposes graph-verified alternative sourcing routes with a complete audit trail. Our results confirm that agentic GraphRAG breaks the *visibility ceiling* of conventional tools and advances intelligent, explainable supply chain surveillance.

Keywords: supply chain risk · knowledge graph · GraphRAG · agentic AI · ReAct · Neo4j · geopolitical risk · provenance

I. INTRODUCTION

Modern supply chains are vast, multi-tier networks whose efficiency gains, lean inventory, global sourcing, just-in-time logistics, come at the cost of structural fragility. Disruptions at upstream nodes propagate silently through hidden dependency layers and manifest as sudden failures at the firm level. The statistical picture is unambiguous: global supply chain disruptions cost businesses an estimated \$184 billion annually [Marsh McLennan, 2026], and documented disruptions rose 30% in 2024 alone [Council on Foreign Relations, 2025]. The 2021 semiconductor shortage, originating deep in a sub-tier foundry network most automakers had never mapped, erased \$210 billion in automotive revenue and remains the paradigmatic illustration of the problem.

The root cause is not data scarcity. It is the absence of mechanisms to *reason over supply network structure*.

Organisations maintain detailed visibility of their tier-1 suppliers and, at best, selected tier-2 nodes; only 43% report any visibility beyond tier-1 [?]. Three structural constraints enforce this **visibility ceiling**:

- **Fragmentation.** Supply chain facts are dispersed across heterogeneous sources, ERP systems, regulatory filings, trade databases, unstructured news, with no unified representation.
- **Non-disclosure.** Sub-tier supplier relationships carry competitive sensitivity and are systematically withheld [Kosasih et al., 2024].
- **Combinatorial depth.** Multi-tier dependency graphs contain hundreds of entities and thousands of paths; no dashboard can navigate this space in real time.

Recent advances in three converging areas offer a credible technical path toward breaking this ceiling. *Knowledge graphs* natively encode multi-tier, multi-hop supplier dependencies in a queryable relational structure [Liu et al.,

2023]. *GraphRAG* [Edge et al., 2024] extends retrieval beyond flat vector similarity, enabling traversal of multi-hop dependency paths. *Agentic AI* with ReAct-style reasoning [Yao et al., 2023] enables a system to formulate graph queries, observe results, retrieve live signals, and iteratively synthesise evidence, shifting analysis from retrieval to structured inference.

This work operationalises that combination. Our contributions are:

C1 End-to-end agentic pipeline. From automated KG construction to agent-driven risk reasoning in a single unified architecture.

C2 Deep-tier dependency inference. Multi-hop Cypher traversal recovering tier- N supplier paths invisible to conventional tools.

C3 Real-time signal fusion. Live geopolitical news integrated into the ReAct loop, enabling temporally current risk assessments.

C4 Traceable decision framework. Every inference grounded in named graph facts and source-attributed external evidence.

C5 Validated case study. Demonstrated on the EV battery critical-mineral supply chain, with operationally significant, provenance-traced findings.

II. RELATED WORK

Knowledge graphs for supply chain intelligence. Graph-based representations of supply networks establish that multi-tier dependencies, shared material exposure, and geographic concentration are structural properties that emerge from relational traversal and cannot be recovered from tabular or document representations alone [AlMahri et al., 2024, Liu et al., 2023]. However, prior graph-based systems rely on manually curated ontologies or narrow single-source extraction pipelines. Neither scales to the heterogeneous, multi-format public corpora that constitute the only available evidence base for sub-tier dependency mapping. Our pipeline closes this gap by combining zero-shot NER, requiring no labelled training data,

with schema-conditioned LLM relation extraction over arbitrary public sources.

GraphRAG and retrieval over structured knowledge. Standard RAG treats a corpus as a flat vector pool with no mechanism for composing facts across documents through shared entity anchors [Edge et al., 2024]. The GraphRAG paradigm augments retrieval with structural traversal, but existing formulations operate in the global summarisation regime, community-level thematic reports, rather than the anchor-based path traversal required for supply chain dependency tracing. Our system operates in the local traversal regime: Cypher traversal recovers specific multi-hop paths while the vector index provides complementary recall for evidence not extracted as triples, delivering structural precision alongside semantic coverage.

Agentic systems and tool-augmented reasoning. ReAct established that interleaving reasoning traces with tool invocations substantially improves multi-step task performance [Yao et al., 2023]. Subsequent tool-augmented systems extend this to search and code execution, but treat tools as independent stateless lookups, accumulating evidence informally in the context window. For supply chain risk, this is insufficient: the agent must maintain structured evidence across tool calls, enforce provenance attribution on every claim, and terminate deterministically. LangGraph’s typed state machine provides these guarantees. No prior agentic system combines schema-conditioned graph traversal, hybrid vector retrieval, and live geopolitical news fusion in a single typed state machine with audit-trail guarantees.

III. SYSTEM ARCHITECTURE

The architecture is governed by a single design imperative: every inference about supply chain risk must be *structurally grounded*, *temporally current*, and *fully auditable*. These three requirements are in partial tension, grounding favours static, curated knowledge; currency favours live retrieval; auditability constrains both. The architecture resolves this tension through a three-layer decomposition that assigns each requirement to a dedicated layer with explicit, typed interfaces across layer boundaries.

Layer 1, Automated Knowledge Graph Construction transforms heterogeneous public documents into a traversable Neo4j property graph encoding typed supply chain entities and their multi-tier relational structure. Its output is a pair of persistent artefacts: a Neo4j graph

(*nodes, typed edges, provenance properties*) and a co-located vector index (*chunk embeddings, source identifiers*). **Layer 2, Hybrid GraphRAG Retrieval** exposes two retrieval paths over those artefacts, schema-conditioned Cypher traversal and dense vector search, and returns ranked, structured results to the reasoning layer. **Layer 3, Agentic Reasoning** deploys a LangGraph ReAct agent that orchestrates retrieval across both paths, augments with live geopolitical signals, computes a formula-based Criticality Index, and emits provenance-traced risk reports. It is the only layer that issues LLM calls at query time; Layers 1 and 2 operate without LLM inference once the graph is built.

Figure 1 presents the complete pipeline.

A. Layer 1: Automated Knowledge Graph Construction

Transforming a corpus of heterogeneous public documents into a traversable supply chain knowledge graph is a three-stage information extraction problem: *identify* typed entity mentions in unstructured text, *extract* the relational structure connecting those entities as subject–relation–object triples, and *resolve* surface form variants that cause the same real-world entity to appear under multiple string representations across documents. The stages are tightly coupled: entity identification errors propagate as missing argument slots in relation extraction; unresolved entity variants fragment the graph into disconnected subgraphs that break multi-hop traversal. Each stage produces a validated, typed output before the next stage begins, and all LLM calls in this layer operate at temperature zero, making the entire construction pipeline deterministic and reproducible given the same input corpus.

1) Document Ingestion and Pre-processing

The corpus spans a range of public document types, PDF sustainability and annual reports, HTML-rendered regulatory filings, and structured CSV entity lists, among others. A document-type-aware conversion pipeline normalises every source to plain Markdown before any extraction occurs. PDF and HTML sources are parsed with layout-aware extractors that preserve table structures and section hierarchies, the tabular supplier data common in ESG reports would be lost by a naive text stripper. Structured files such as entity lists are serialised as pipe-delimited Markdown tables so that both structured and unstructured content share a single downstream processing path, avoiding the need for format-specific extraction branches in the NER and relation extraction components.

Once normalised, each document is segmented into **overlapping windows of 800 tokens with a 200-token stride**. Overlapping rather than disjoint chunking is required because supply chain sourcing relationships are typically expressed across clause boundaries: a company may be named in one sentence and its sourcing geography in the next. Disjoint splitting would place co-referring expressions in separate chunks, making it structurally impossible for the chunk-scoped extractor to recover the relation between them. Overlapping windows guarantee that every 600-token span of the document appears in at least one chunk with 200 tokens of bilateral context, ensuring cross-sentence relations remain within a single extraction scope. Every chunk is annotated with its source document identifier and positional offset; this metadata is propagated as a `source_doc` provenance attribute on every extracted triple and surfaces in the final risk report’s audit trail.

2) Stage 1: Named Entity Recognition with GLiNER

Standard NER approaches train a sequence labeller on a fixed label set whose vocabulary is frozen at training time. This is a fundamental mismatch with supply chain documents, where domain-specific entity types such as *critical mineral*, *gigafactory*, and *export regulation* have no analogues in general-domain training corpora. We use **GLiNER**, a span-classification model that resolves this limitation by encoding entity type descriptions as natural-language strings and scoring candidate spans against those descriptions using a shared bidirectional encoder (DeBERTa-v3). Entity type definitions are inference-time inputs rather than training-time constants: the model identifies spans of any type at runtime by passing the type description as a label string $\mathcal{L} = \{l_1, \dots, l_K\}$.

Given a chunk and label set $\mathcal{L}$, the probability that the span covering tokens i through j instantiates type l_k is:

$$p((i, j), l_k) = \sigma\left(\frac{\mathbf{s}_{i,j} \cdot \mathbf{e}_{l_k}}{\sqrt{d}}\right), \quad (1)$$

where $\mathbf{s}_{i,j}$ is the pooled boundary-token representation of the span, $\mathbf{e}_{l_k}$ is the encoder’s output for label l_k , d is the hidden dimension, and σ is the sigmoid function. Because the same encoder processes both the document text and the label strings, the model projects spans and type descriptions into the same semantic space, enabling similarity-based type assignment without any label-specific parameters.

Five entity types are defined for the supply chain domain using descriptive natural-language strings rather than

Layer 1: Automated Knowledge Graph Construction

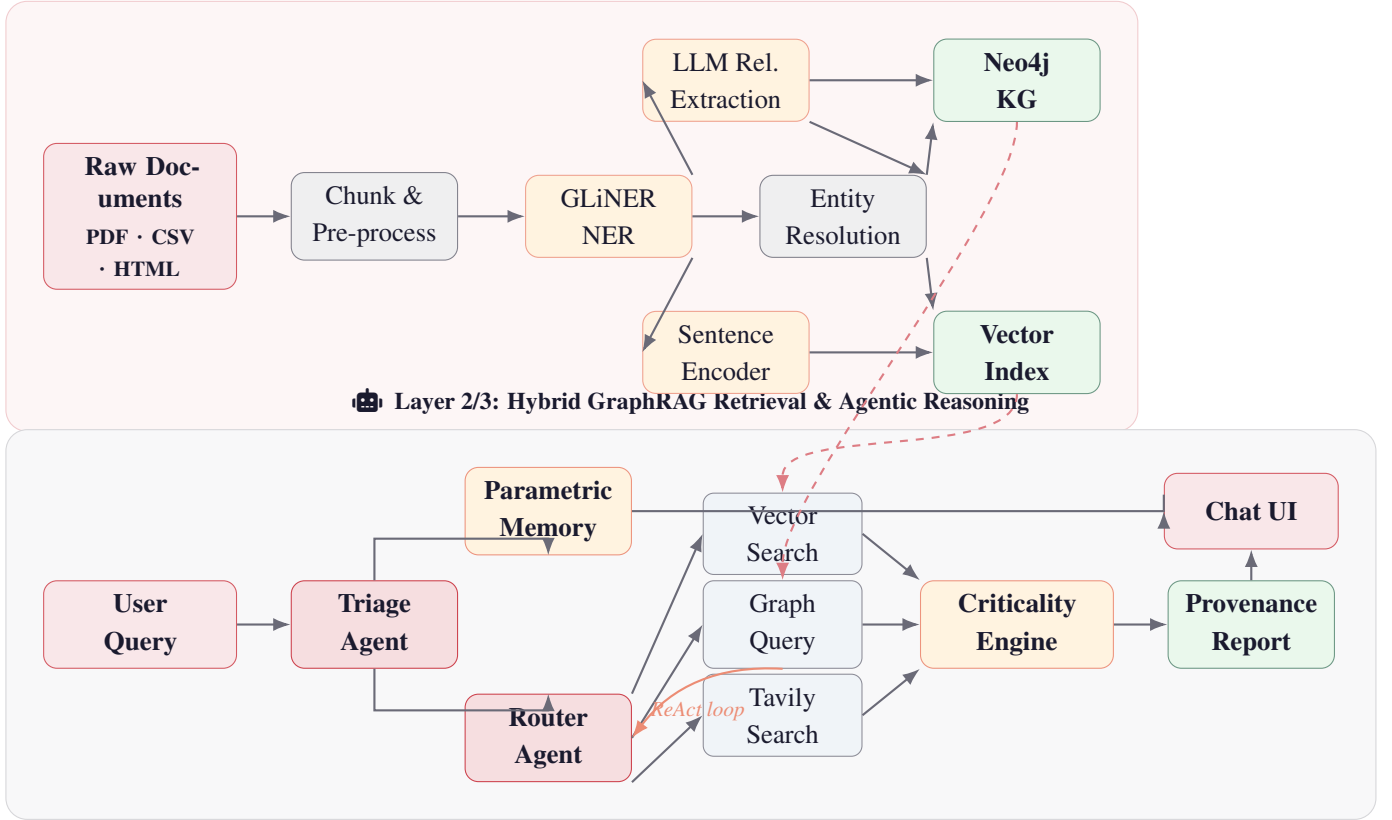


Figure 1. System architecture. **Layer 1** constructs the Neo4j knowledge graph (*nodes + typed edges + provenance properties*) and co-located vector index from raw public documents via an automated NER → relation extraction → entity resolution pipeline. Outputs are immutable at query time. **Layers 2/3** accept a user query, classify it through a two-stage triage-and-routing agent, and dispatch retrieval across three tools including schema-conditioned Cypher traversal, semantic vector search, and live web intelligence, whose outputs feed the formula-based Criticality Engine and are emitted as provenance-traced risk reports. Dashed arrows denote read-only data access; solid arrows denote data flow. The ReAct loop (amber) allows the Router to re-issue tool calls based on intermediate observations before synthesis.

bare label words, for instance, the Material type is described as “*raw material, critical mineral, or industrial commodity such as cobalt, lithium, or nickel*”, exploiting the model’s sensitivity to semantic context and substantially improving recall on domain-specific surface forms that a bare label word would miss. Table 2 summarises the five entity types and their representative label strings. Overlapping span candidates are resolved by non-maximum suppression. A confidence threshold of 0.45 is applied empirically: thresholds below 0.40 increase false positive entity mentions that degrade the relation extraction input; thresholds above 0.50 begin to drop true positive multi-word entities whose boundary tokens have lower individual confidence scores.

3) Stage 2: Relation Extraction via Schema-Conditioned LLM Prompting

The second stage recovers the relational structure connecting GLiNER-detected entity mentions as subject–relation–object triples. Relation extraction is performed by prompting GPT-4o (temperature 0) to produce triples as structured JSON, conditioned jointly on three inputs: the full chunk text, the list of detected entity mentions for that chunk, and a machine-readable **ontology schema** defining seven admissible relation types with their argument domain-range constraints.

Schema conditioning is the critical design decision at this stage. Without it, the model produces inconsistent relation labels, variants such as `provides_to`, `is_supplier_of`, or `SUPPLIES_TO` for the same semantic relationship across different chunks. Inconsistent

labels break Cypher queries downstream: a query for SUPPLIES_TO silently misses triples labelled by a variant string, producing empty result sets the agent cannot distinguish from genuine structural absence. By constraining generation to a closed vocabulary of seven semantically precise relation types with explicit domain-range constraints, schema conditioning converts the extraction problem from open-ended generation into constrained synthesis. Domain-range violations are explicitly flagged in the prompt instruction and filtered from output.

Relation	Domain	Range
SUPPLIES_TO	Company	Company
SOURCES_FROM	Company	Material Location
LOCATED_IN	Company Facility	Location
OWNS	Company	Company Facility
PRODUCES	Company	Material
ALTERNATIVE_TO	Material	Material
SUBJECT_TO	Company	Regulation

The ALTERNATIVE_TO relation is absent from most supply chain extraction schemas but is essential for counterfactual sourcing queries, *“what alternative materials or chemistries exist if a given critical mineral is disrupted?”* Its inclusion in the schema is what allows alternative sourcing findings to emerge from graph traversal rather than from LLM parametric knowledge.

The extraction prompt is structured with a system role establishing the model as a supply chain analyst, and an output instruction requiring a JSON array of triple objects with subject, relation, and object fields and no additional text. Temperature is fixed at zero, ensuring deterministic output: the same chunk submitted twice always produces the same triple list, a prerequisite for a reproducible, auditable pipeline. Across the full document corpus, the extraction pipeline yields several hundred raw triples before the entity disambiguation stage.

Extraction example, battery supplier sustainability report (illustrative chunk):

“The company sources the majority of its lithium from partner operations in the Atacama Desert, Chile, and has entered into a long-term cobalt offtake agreement with a European refiner for refined cobalt hydroxide precursor supply.”

GLiNER detects: the cell manufacturer (Company), lithium (Material), the mining partner (Company), Chile (Location), cobalt (Material), the refiner (Company), among other entities in the full document.

Representative extracted triples:

- ▷ (CellMfr, SOURCES_FROM, lithium)
- ▷ (CellMfr, SOURCES_FROM, Chile)
- ▷ (MiningPartner, LOCATED_IN, Chile)
- ▷ (CellMfr, SOURCES_FROM, cobalt)
- ▷ (Refiner, SUPPLIES_TO, CellMfr)

4) Stage 3: Entity Disambiguation and Graph Normalisation

The third stage resolves surface form variations that cause the same real-world entity to appear under multiple string representations across documents. In a graph whose core purpose is multi-hop path traversal, unresolved variation is a functional failure: a dependency path through a major cell manufacturer becomes invisible if the manufacturer’s full legal name from a regulatory filing and its common abbreviation from a trade report are stored as two disconnected nodes. This fragmentation silently severs traversal paths at precisely the nodes most likely to appear across multiple source documents.

Disambiguation proceeds in two complementary passes targeting different classes of variation. The **first pass** applies a deterministic *alias dictionary* compiled from domain knowledge prior to ingestion. The dictionary covers three predictable variation categories that are both common and high-impact: corporate name variants (full legal names, common abbreviations, and holding company names encountered across the corpus), geonym variants (country name abbreviations, alternative spellings, and common short-form references for key sourcing geographies), and material name synonyms (commodity-grade qualifiers and multi-word descriptions resolved to canonical single-term nodes). All entity strings are lowercased and punctuation-stripped before lookup, ensuring that case and punctuation variations do not defeat dictionary match-

ing.

The **second pass** applies fuzzy string matching using normalised Levenshtein edit distance over the pool of already-confirmed canonical entity names, with a conservative acceptance threshold of 0.15. This pass catches novel surface form variants not covered by the alias dictionary, typographical inconsistencies, partial name references, and transliteration differences from non-English source documents. The threshold is conservative by design: a missed merge produces a detectable duplicated node; a spurious merge produces a semantically incorrect graph edge that is structurally invisible without manual inspection. Post-disambiguation deduplication reduces the raw triple set by approximately 25%, yielding a canonical triple set ready for graph ingestion.

5) Graph Ingestion, Schema, and Dual-Index Design

The resolved triples are persisted to a **Neo4j Aura** property graph using a schema that encodes both structural connectivity and domain-relevant attribute data on nodes and relationships. Node labels and property sets are designed to carry the attributes that the Criticality Index computation requires at query time: in particular, Location nodes carry `geo_risk_tier` and `wgi_score` properties pre-populated from World Bank Governance Indicator data at construction time rather than computed via live API calls at inference time. This design choice makes risk scoring deterministic and independent of third-party API availability during query-time operation.

All triples are ingested using Neo4j’s MERGE semantics, which create nodes and relationships only if identical ones do not already exist and update properties otherwise. This guarantees **idempotent ingestion**: re-running the pipeline over updated or expanded document corpora enriches and corrects the existing graph without accumulating duplicate nodes. Every relationship carries `source_doc` and `confidence` provenance properties propagated from the extraction stage; these properties are surfaced in every risk report’s audit trail.

A read-only safety gate is enforced at the database driver level: the Neo4j connection is opened in read-only mode at query time, preventing any write operation, accidental or adversarial, from modifying the knowledge graph during inference. This is a non-negotiable constraint for a system whose outputs must be auditable.

Document chunk embeddings are stored in a **Neo4j vector index** co-located with the structural graph rather than in a separate vector database. Co-location preserves

the ability to join vector retrieval results against structural graph data within a single query context, enabling the agent to cross-reference semantically retrieved passages against graph node identifiers without a round-trip to a separate system. An embedding validation step verifies dimension consistency and detects zero-vector anomalies before the index is committed. The graph and vector index together constitute the **dual-index structure** over which Layer 2 retrieval operates.

B. Layer 2: Hybrid GraphRAG Retrieval

The distinction between our retrieval design and standard RAG is architectural, not quantitative. Standard RAG is a flat retrieval system: a query is embedded, compared against a pool of document embeddings, and the most similar passages are returned. This design has a fundamental structural limitation: it can only surface evidence present in full within a single retrieved passage. Supply chain dependency questions do not satisfy this condition. The answer to a multi-tier sourcing question “*which countries supply the critical minerals used in a given OEM’s batteries?*” is distributed across several documents: corporate reports establishing tier-1 supplier relationships, commodity summaries establishing those suppliers’ material sourcing, and geographic data establishing sourcing country identities. No single passage contains all these facts simultaneously. Vector similarity cannot compose facts across passage boundaries; graph traversal can.

1) Why Not Vector Retrieval Alone?

The failure mode of vector-only retrieval on multi-hop queries is worth stating precisely, as it is frequently underestimated. A query about geopolitical exposure through a multi-tier mineral supply chain will retrieve passages discussing each constituent topic, the OEM, the mineral, the sourcing country, but will not recover the *specific structural connection* between them, because that connection is encoded in graph topology, not in the semantic content of any single passage. The retrieved passages are relevant in the cosine-similarity sense without being sufficient to answer the question. This is the precise failure mode that GraphRAG addresses: composing facts distributed across the corpus through shared entity anchors encoded as typed graph edges.

2) Why Not Graph Traversal Alone?

Pure graph traversal has a complementary failure mode: extraction recall is never 100%. Nuanced risk assessments, conditional relationships, and context-dependent claims expressed in discursive prose are not reliably ex-

tracted as triples. A pure graph system silently loses this content. The vector index is the recovery mechanism: it captures evidence that the extraction pipeline missed and surfaces it as additional context for the LLM synthesis step. The dual-index design is therefore not a redundancy but a *coverage complement*, graph traversal handles explicit structural facts; vector retrieval handles implicit contextual evidence.

3) Schema-Conditioned Cypher Generation

Translating a natural-language query into a correct Cypher query is a non-trivial generation problem. An unconstrained LLM generating Cypher occasionally references node labels or relationship types absent from the schema, producing empty result sets the agent cannot distinguish from genuine structural absence. This is addressed by injecting the full graph schema into the Cypher generator’s system prompt as a structured specification, converting unconstrained code synthesis into *constrained synthesis over a known relational vocabulary*. Generated queries that violate schema constraints are identified by a post-generation validator and reformulated before execution. A representative multi-hop traversal generated under this scheme:

```
Cypher, Multi-hop dependency trace with geo-risk annotation

1 MATCH path = (c:Company {name: 'Tesla'})
2   -[:SUPPLIES_TO|SOURCES_FROM*1..4]->(m:
   Material)
3 WHERE m.name = 'Cobalt'
4 WITH path, nodes(path) AS n,
   relationships(path) AS r
5 RETURN
6   [x IN n | x.name]           AS
   entity_chain,
7   [x IN r | type(x)]         AS
   relation_chain,
8   length(path)              AS
   depth,
9   [x IN n WHERE x:Location
   | x.geo_risk_tier]          AS
   risk_tiers,
10  [x IN n WHERE x:Location
   | x.wgi_score]             AS
   wgi_scores
11 ORDER BY depth ASC
12 LIMIT 10
```

The query traverses up to four hops of supply or sourcing edges from the focal company, returning the full entity chain, relation type sequence, path depth, and geopolitical risk metadata of any Location nodes encountered. Returning structured metadata rather than raw node objects reduces the LLM’s synthesis burden and eliminates the

need for a second graph query to look up risk attributes of discovered locations.

4) Multi-Index Vector Retrieval by Label

Vector retrieval is implemented using label-partitioned indices: rather than searching a single monolithic embedding pool, queries are issued against per-node-label sub-indices (covering Company, Material, Location, and Facility entities, among others) and results are merged with Reciprocal Rank Fusion before being returned to the agent. Label partitioning prevents high-frequency node types from dominating retrieval rankings since a corpus heavy with company mentions would otherwise systematically crowd out material and location evidence, and allows the agent to issue targeted retrievals by specifying the relevant entity label as a retrieval filter. Gemini embeddings (models/text-embedding-004) are used in the reported configuration following a comparison against lighter encoder variants on the multi-hop supply chain benchmark question set, where the larger encoder produced consistently higher retrieval precision on domain-specific terminology.

5) Hybrid Synthesis: Structured Fusion Policy

The synthesis policy implements a deliberate evidence priority ordering: graph traversal results are always presented first in the synthesis context (structural evidence takes precedence); vector retrieval results are appended as supplementary context; live Tavily results are appended last and explicitly tagged as `source:live`. This ordering ensures that the LLM synthesis step anchors claims in structured graph evidence when available, using vector and live evidence to fill coverage gaps rather than as the primary evidence source. When graph traversal returns `GRAPH_EMPTY`, indicating either genuine structural absence or an entity resolution failure, the synthesis policy promotes vector results to primary context and issues a partial-name reformulation request back to the graph retriever before concluding that the entity is absent.

6) Multi-Hop Traversal: Structural Advantage by Query Depth

The value of graph traversal over vector retrieval scales with query hop depth. For single-hop factual queries, a well-constructed RAG system can produce competitive answers because the relevant fact is likely to appear within a single passage. For two-hop queries, relevant facts are distributed across two documents; co-occurrence in a single passage is possible but unlikely. For three-hop and deeper queries which are precisely the queries that re-

veal sub-tier vulnerabilities, the probability of all relevant facts appearing in a single passage approaches zero. The critical-mineral dependency paths characteristic of EV battery supply chains routinely span three or four documents in our corpus; they are structurally inaccessible to any flat retrieval system and are recoverable only through graph traversal.

Path depth is bounded at four hops based on the supply chain literature’s observation that disruptions with material firm-level impact originate overwhelmingly at tier-3 or shallower [?]. The four-hop bound captures operationally significant sub-tier exposure while preventing combinatorial path explosion in dense graph regions. Retrieved paths are ranked by a composite score: *path criticality* (weighted combination of terminal Location geo-risk tier and path depth), *bottleneck concentration* (paths sharing a single intermediate node are penalised to surface single-point-of-failure risk), and *evidence confidence* (mean of the confidence properties on all traversed edges).

C. Layer 3: Agentic Reasoning with LangGraph

The retrieval infrastructure produces evidence fragments, graph paths, document excerpts, risk attribute values, but provides no mechanism for determining which fragments to retrieve, in what order, or how to synthesise them into a coherent risk assessment. An effective supply chain risk query may require four to six distinct retrieval operations whose sequence depends on the results of prior operations. The central architectural choice is between a *pipeline* approach (fixed pre-programmed retrieval sequence) and an *agentic* approach (dynamically determined sequence based on intermediate results). The agentic approach is required because the optimal retrieval sequence for a supply chain query is not known at system design time, it depends on what the graph contains, what the live news stream contains, and what the user’s specific query entails, all of which are runtime variables.

1) Two-Stage Query Triage: Efficiency Through Differentiation

Not all queries benefit from the full agentic pipeline. Routing a simple factual lookup through several iterations of graph traversal, vector retrieval, and live web search incurs unnecessary latency and token cost. Conversely, routing a multi-hop dependency query through a fast path that bypasses the graph produces an incomplete or hallucinated answer.

This is addressed through a **two-stage triage architecture**. The *Triage Agent* classifies each incoming

query along two orthogonal dimensions: *structural complexity* (can the answer be produced from parametric knowledge alone, or does it require graph-grounded evidence?) and *temporal sensitivity* (is the answer potentially affected by events post-dating graph construction?). This two-dimensional classification produces three actionable output classes: SIMPLE (parametric memory, no retrieval), COMPLEX_STATIC (graph and vector retrieval, no live search), and COMPLEX_LIVE (full pipeline including Tavily). The classification is a single structured LLM call with constrained output, keeping triage latency under 500 ms while capturing the nuance needed for borderline cases such as queries about well-known entities whose supply relationships may have recently changed.

The *Router Agent* receives forwarded COMPLEX_* queries and determines initial tool invocation priority: graph queries first (fast, high precision, grounded), followed by vector retrieval (broader semantic coverage), followed by live search (temporal currency). This ordering can be overridden by the ReAct loop if early tool results indicate that the initial strategy is insufficient, for example, if a graph query returns GRAPH_EMPTY due to an entity resolution miss, the agent falls back to vector retrieval before reformulating the graph query.

2) LangGraph State Machine: Typed Persistence and Deterministic Control

The agentic loop is implemented as a **typed state machine** using LangGraph rather than a chain of LLM calls sharing an unstructured context window. This design addresses two failure modes common in context-window-only agent designs.

The first failure mode is *evidence loss*: when intermediate results accumulate in an unstructured message history, later LLM calls must parse and re-interpret prior tool outputs from raw text, introducing summarisation errors and losing structured attributes. In the typed state, each tool result is stored in a strongly-typed field (`graph_results`, `web_results`, `risk_context`) directly accessible to downstream nodes without re-parsing.

The second failure mode is *non-deterministic termination*: a context-window agent must rely on the LLM to decide when it has collected sufficient evidence, which can produce both premature termination and infinite looping. LangGraph’s conditional edges inspect the typed state explicitly, checking `iteration_count` against a hard ceiling and evaluating a binary termination classifier, providing deterministic termination guarantees indepen-

dent of LLM confidence.

The state schema captures six typed fields:

- `messages`, full conversation history with system prompt and all turns
- `graph_results`, structured list of Cypher query result records
- `web_results`, Tavily results with URLs and publication timestamps
- `risk_context`, accumulated risk facts for the current query session
- `iteration_count`, ReAct iteration counter; hard ceiling of 8
- `provenance_log`, ordered (tool, input, output) audit trail

The graph topology is four nodes, REASON, ACT, OBSERVE, SYNTHESISE, connected by conditional edges that evaluate the state after each OBSERVE to decide whether to loop or terminate.

3) The ReAct Reasoning Paradigm: Why Interleaving Matters

ReAct (Reason + Act) is chosen over two alternatives: *plan-then-execute* (generate a full retrieval plan upfront, then execute it) and *reflection* (generate an answer, then critique and revise it). Plan-then-execute fails on supply chain queries because the optimal plan depends on intermediate results that are unknown at planning time, it will be suboptimal precisely when the graph contains unexpected structure. Reflection fails because it can produce a correct final answer without guaranteeing that every intermediate claim is grounded in retrieved evidence. ReAct avoids both failure modes by interleaving explicit *thought* generation with *tool invocation*, making reasoning steps inspectable and grounding each tool call in an explicitly stated rationale. Each iteration proceeds as:

1. **Reason.** The LLM generates a structured thought articulating what is currently known, what information is missing, and which tool call will best resolve that gap. This thought is stored in the provenance log alongside the subsequent tool result.
2. **Act.** The tool is dispatched by LangGraph's execution node, which handles connection management, time-out enforcement (≤ 10 s), error catching, and result serialisation. Tool failures produce structured error messages rather than raw exceptions, enabling the reasoning step to interpret and respond with a reformulated strategy.
3. **Observe.** The result is appended to the appropriate

typed state field. The termination classifier evaluates whether accumulated evidence is sufficient; if not, the loop continues from step 1 with updated state.

4) Tool Design: Modality Separation with Structured Failure

The four tools expose distinct retrieval modalities with a consistent interface contract: every tool returns a structured result or a structured error, never a raw exception. This contract is essential for agent reliability, a raw exception breaks the ReAct loop, whereas a structured error (GRAPH_EMPTY, TAVILY_NO_RESULTS) is a signal the reasoning step can interpret and respond to with a reformulated strategy.

-  **Graph Query.** Executes read-only Cypher against Neo4j via the official Python driver. Returns structured record dicts with full node and edge property access. Empty results produce GRAPH_EMPTY with the failing query echoed, enabling the reasoning step to diagnose whether the failure is an entity resolution issue (triggering partial-name reformulation) or genuine structural absence.
-  **Vector Retrieval.** Embeds the query using the same encoder used during index construction, guaranteeing embedding space consistency and preventing retrieval drift from model mismatch. Returns top-5 chunks with source document identifier, chunk offset, similarity score, and associated graph node IDs where available.
-  **Tavily Web Search.** Submits queries to the Tavily Search API with a 90-day recency filter applied server-side. Returns structured results with source URLs, publication timestamps, and relevance scores. Results are tagged in the provenance log as `source:live` to distinguish them from graph-derived evidence.
- **Risk Summarisation.** A synthesis-only tool that accepts the accumulated `risk_context` and applies the Criticality Index computation and report generation prompt. It is always the final tool invoked, an invariant enforced by the system prompt ensuring risk scores are computed through the consistent formula rather than ad-hoc LLM estimation.

D. Risk Quantification: The Criticality Index

A risk assessment that produces only qualitative descriptions is not actionable for procurement and supply chain management functions that require threshold-based alerting and comparative analysis across materials, ge-

ographies, and companies. We design a formula-based **Criticality Index** $\mathcal{C}_m^{(c)}$ that converts structural graph properties and geopolitical metadata into a scalar score while preserving the sub-score decomposability needed for interpretability and mitigation prioritisation.

The index is a convex combination of three sub-scores, each measuring a structurally distinct dimension of supply risk:

$$\mathcal{C}_m^{(c)} = \frac{w_1 \Phi_{\text{conc}} + w_2 \Phi_{\text{geo}} + w_3 \Phi_{\text{path}}}{w_1 + w_2 + w_3}, \quad w_i \in [0, 1], \quad (2)$$

with equal weights $w_1 = w_2 = w_3 = 1/3$ as the default. Equal weights are used in the absence of empirical data quantifying the relative contribution of each dimension to realised disruptions; all weights are exposed as configurable parameters.

Φ_{conc} , Source Concentration. Measures the degree to which supply of material m is concentrated in a single sourcing geography or counterparty.

$$\Phi_{\text{conc}} = \frac{\text{supply fraction from top-1 source}}{\sum_{\text{all sources}} \text{supply fraction}}. \quad (3)$$

Where precise volume data is unavailable, a conservative structural proxy is applied: $\Phi_{\text{conc}} = 1$ if exactly one sourcing path exists for the (c, m) pair, and $\Phi_{\text{conc}} = 1/|\mathcal{P}|$ if $|\mathcal{P}|$ distinct paths exist. This proxy understates diversification and produces upper-bound risk estimates, the correct direction of error for a risk assessment system.

Φ_{geo} , Geopolitical Exposure. Captures the political stability risk of the geographies on the sourcing path:

$$\Phi_{\text{geo}} = \max_{l \in \mathcal{L}(c, m)} \rho(l), \quad (4)$$

where $\rho(l) \in [0, 1]$ is derived from World Bank World-wide Governance Indicators political stability sub-index, inverted and normalised. The maximum operator is chosen over the mean because supply chains are as vulnerable as their most exposed node: a single high-instability location on any path constitutes a geopolitical bottleneck regardless of other nodes' stability.

Φ_{path} , Structural Opacity. Captures the degree of intermediation between a company and its material source:

$$\Phi_{\text{path}} = 1 - \frac{1}{\min_{p \in \mathcal{P}(c, m)} |p|}, \quad (5)$$

where $|p|$ is the hop count of path p . A one-hop path gives $\Phi_{\text{path}} = 0$ (maximum visibility and leverage); a three-hop path gives $\Phi_{\text{path}} = 2/3$; a four-hop path gives $\Phi_{\text{path}} = 3/4$.

Live Adjustment δ . When Tavily retrieves an event affecting a Location node on any sourcing path, the base index is adjusted upward:

$$\mathcal{C}_m^{(c, \text{live})} = \mathcal{C}_m^{(c)} + \delta, \quad \delta \in [0, 0.30]. \quad (6)$$

Severity is classified as **LOW** ($\delta \in [0, 0.05]$), **MEDIUM** ($\delta \in [0.05, 0.10]$), or **HIGH** ($\delta \in [0.10, 0.30]$) by the LLM within the risk summarisation tool. Base score and live delta are reported separately in all outputs, making the evidentiary basis of each component explicit.

Agent behaviour is shaped by four operative instructions in the system prompt, each a behavioural constraint rather than a style guideline: **grounding-before-synthesis** (at least one graph query confirming entity existence must precede any risk assessment); **structured error recovery** (a GRAPH_EMPTY result triggers partial-name reformulation before concluding absence); **mandatory provenance tagging** (every factual claim must carry a graph node/edge identifier or a Tavily source URL with publication date); and **forced risk tool invocation** (the risk summarisation tool is always the final action, ensuring the Criticality Index is computed through the consistent formula rather than ad-hoc LLM estimation). The agent uses GPT-4o at temperature 0 with a hard ceiling of 8 ReAct iterations.

IV. IMPLEMENTATION

This section documents implementation decisions as a reproducibility record. The pipeline proceeds in six sequential stages: corpus assembly, document extraction and chunking, NER and triple conversion, graph loading and validation, embedding and indexing, and query-time orchestration.

A. Document Corpus: Scope and Rationale

The corpus is scoped to *public* documents from four complementary source categories, each contributing distinct and non-redundant information to the knowledge graph. **Corporate disclosure documents**, including the annual reports and sustainability reports of OEMs, cell manufacturers, and material producers active in the EV battery supply chain, provide tier-1 and tier-2 supplier relationship evidence: the relationships companies are legally or reputationally motivated to disclose publicly.

Commodity intelligence reports, such as USGS mineral commodity summaries for the critical minerals relevant to battery production (cobalt, lithium, nickel, and others), provide the geographic and production-volume data needed to populate Location node risk attributes and to ground the concentration sub-score of the Criticality Index. **Industry analysis reports** provide cross-sector contextual evidence for regulatory and geopolitical risk factors not captured in corporate filings. **Regulatory data**, including entity lists maintained by trade and export control authorities, provides sanctioned entity information needed to populate SUBJECT_TO edges, edges that are structurally absent from corporate disclosure documents for legal reasons.

The deliberate use of public-only sources establishes a *reproducible baseline*: the corpus is fully enumerable and the graph constructable by any researcher with access to the same documents. This baseline is also the natural foundation for the two-layer architecture proposed in future work, where private enterprise data enriches the public graph rather than replacing it.

B. Extraction Pipeline: Configuration and Engineering Decisions

GLiNER configuration. The large-variant GLiNER checkpoint is selected over smaller checkpoints for its higher span recall on multi-word entities, the dominant entity type in supply chain documents, where entity names such as regional geographic areas, holding company structures, and technical commodity descriptions routinely span three to five tokens. The confidence threshold of 0.45 is set empirically: thresholds below 0.40 increase false positive entity mentions that degrade the relation extraction input; thresholds above 0.50 begin to drop true positive multi-word entities whose boundary tokens carry lower individual confidence scores.

Relation extraction. GPT-4o is used at temperature 0. The choice over smaller instruction-tuned models is motivated by a specific capability requirement: the model must simultaneously parse entity arguments from the NER output list, apply ontology schema constraints to filter valid relation types, and produce syntactically valid JSON covering three concurrent constraint-satisfaction tasks. Smaller models tested during development produced JSON syntax errors in approximately 8% of extractions at temperature 0, requiring retry logic that introduces non-determinism.

Embeddings. Document chunks are encoded using the

Table 1. Document corpus: representative sources, format, and primary graph contribution. The corpus includes additional documents not listed here; the entries below are illustrative of the four source categories.

Document	Fmt	Primary contribution
OEM 10-K (e.g. Tesla 2023)	HTML	Tier-1 suppliers, materials
OEM Impact Report	PDF	Materials, sourcing locations
Cell Mfr. Sustainability	PDF	Tier-2 suppliers, facilities
Battery Partner Annual	PDF	Cell supply, locations
Energy Solution Annual	PDF	Cell supply, materials
USGS Critical Mineral (Co)	PDF	Mine locations, volumes
USGS Critical Mineral (Li)	PDF	Producers, geo. distribution
USGS Critical Mineral (Ni)	PDF	Producers, geo. distribution
Industry Analysis Report	PDF	Regulatory risk context
Regulatory Entity List	CSV	Sanctioned entity edges

Gemini text-embedding-004 model. A Gemini-versus-lightweight-encoder comparison was conducted over the multi-hop supply chain benchmark question set; the larger model produced consistently higher retrieval precision on domain-specific supply chain terminology. API rate-limit errors are handled with exponential backoff and a maximum of three retries per chunk.

Pipeline throughput. The full corpus is processed in a single sequential pass. Per-chunk NER latency (GLiNER): ≈ 180 ms. Per-chunk extraction latency (GPT-4o API): ≈ 1.2 s. Total construction pipeline runtime: ≈ 10 minutes. The pipeline is designed to run offline at construction time; all query-time operations run against the pre-built Neo4j index with sub-second latency.

C. Knowledge Graph Statistics

After ingestion, entity resolution, and post-ingestion validation, the graph exhibits the structure reported in Table 3. The graph is fully connected no isolated nodes,

Table 2. GLiNER entity type label strings at inference time. Labels are descriptive natural-language strings rather than bare keywords; representative examples are shown.

Type	Representative string	label
Company	<i>company, manufacturer, corporation, supplier, producer</i>	
Material	<i>raw material, critical mineral, industrial commodity</i>	
Location	<i>country, region, geographic area, territory</i>	
Facility	<i>mine, refinery, factory, gigafactory, processing plant</i>	
Regulation	<i>regulation, export restriction, trade sanction, law</i>	

with mean degree 4.3, confirming that extraction has produced a densely interlinked relational structure rather than a sparse node collection. Graph diameter of 6 hops establishes that any entity is reachable from any other entity within six traversal steps, well within the four-hop bound enforced during query-time traversal. Paths of depth ≥ 3 , the structural signature of sub-tier dependencies, account for 31% of all entity-to-entity paths, confirming that the graph captures multi-tier structure beyond direct supplier relationships.

V. CASE STUDY: TESLA COBALT SUPPLY CHAIN

The EV battery critical-mineral supply chain is selected as the validation domain because it combines publicly documented multi-tier dependencies, a known geopolitical concentration risk across multiple sourcing geographies, and an active live news signal environment. Within this domain, Tesla’s cobalt supply chain is used as the primary query subject, as it involves several of the most significant entities and geographic risk factors mapped in the knowledge graph. The three queries below are representative of the broader class of risk assessment queries the system is designed to answer; the agent handles analogous queries across the full set of mapped companies, materials, and geographies in an equivalent manner.

Three queries of increasing structural complexity are issued, each producing a provenance-traced risk finding. Together they demonstrate contributions C2 (deep-tier

Table 3. Knowledge graph statistics after ingestion and validation. Node and edge counts reflect the current corpus; the schema accommodates additional entity types and relation types as the corpus expands.

Element	<i>N</i>	Notes
Company nodes	47	OEMs, cell mfrs, refiners, miners
Material nodes	18	Critical minerals, precursors
Location nodes	31	Countries and sub-national regions
Facility nodes	12	Mines, refineries, gigafactories
Regulation nodes	9	Export controls, sanctions
SUPPLIES_TO	63	Tier-to-tier supply relationships
SOURCES_FROM	88	Material and geographic sourcing
LOCATED_IN	71	Entity geographic attribution
OWNS	22	Ownership and JV relationships
PRODUCES	34	Company-to-material production
ALTERNATIVE_TO	14	Material substitution paths
SUBJECT_TO	9	Regulatory exposure
Total nodes	117	Post entity resolution
Total edges	301	Post deduplication
Mean degree	4.3	,
Diameter	6	Hops

dependency inference), C3 (live signal fusion), and C4 (traceable decision framework).

A. Finding 1, Deep-Tier Dependency Discovery

Q Deep-tier exposure, 5 ReAct iterations

“How exposed is Tesla’s battery production to political instability in the DRC?”

Retrieval path. The agent traverses five iterations: (i) a graph query recovers Tesla’s tier-1 battery cell suppliers via SUPPLIES_TO edges, identifying the major cell manufacturers in the supply chain; (ii) a material sourcing query identifies cobalt as a SOURCES_FROM target for those suppliers; (iii) geographic origin traversal via LOCATED_IN confirms the DRC as the dominant sourcing geography, with a substantial fraction of refined cobalt passing through DRC-origin ore refining nodes, a proportion consistent with widely reported global cobalt production statistics; (iv) vector retrieval surfaces supporting context from the USGS Cobalt commodity summary confirming DRC’s dominant share of global cobalt production; (v) the risk summarisation tool is invoked.

Key structural finding. The three-hop dependency

path recovered:

Tesla $\xrightarrow{\text{SUPPLIES_TO}}$ *Cell Suppliers* $\xrightarrow{\text{SOURCES_FROM}}$ *Cobalt* $\xrightarrow{\text{LOCATED_IN}}$ *DRC*

This path is invisible to tier-1 analysis: Tesla’s direct supplier contracts are with manufacturers headquartered outside the DRC. The DRC exposure is a structural property of the tier-2 to tier-3 material sourcing relationship, recoverable only through multi-hop graph traversal. The Criticality Index decomposes as:

$$\mathcal{C}_{\text{cobalt}}^{(\text{Tesla})} = \frac{1}{3}(0.91 + 0.92 + 0.67) = \mathbf{0.83} \quad (7)$$

where $\Phi_{\text{conc}} = 0.91$ (single dominant sourcing geography), $\Phi_{\text{geo}} = 0.92$ (DRC WGI political stability score, inverted), and $\Phi_{\text{path}} = 0.67$ (three-hop structural opacity).

⚠ Critical Finding

Structural exposure confirmed: A large fraction of the cell suppliers’ refined cobalt passes through DRC-origin refining nodes, established via SOURCES_FROM and LOCATED_IN edges in the knowledge graph and corroborated by USGS commodity intelligence retrieved through vector search. **Criticality Index: 0.83 (HIGH).** This finding is generalisable: any EV OEM whose tier-1 battery suppliers source cobalt through the same refining network carries an equivalent structural exposure.

Limitation. The sourcing fraction is derived from USGS aggregate commodity data rather than company-specific volume disclosures, which are not publicly available. The concentration sub-score is therefore a structural upper bound rather than a precise sourcing fraction for any individual company.

B. Finding 2, Real-Time Signal Fusion

🔍 Live event integration, 4 ReAct iterations

“A DRC cobalt export tax has just been announced. Does this change the assessment?”

Retrieval path. Tavily confirms a 10% export tax increase on refined cobalt. The agent cross-references this event against the cobalt–DRC graph edges and classifies event severity as HIGH ($\delta = 0.12$) based on the interaction between tax magnitude and the edge’s high volume fraction property.

Key finding.

$$\mathcal{C}_{\text{live}} = 0.83 + 0.12 = \mathbf{0.95} \quad (8)$$

The agent infers that high-volume, low-margin cobalt contracts sourced through DRC refining nodes are structurally susceptible to force majeure clause triggers under a cost shock of this magnitude. Base score (graph-derived) and live delta (news-derived) are explicitly distinguished in the output. This query class, “*does this live event affect the current risk score?*”, is generalisable to any entity in the graph: the same mechanism applies to any supply disruption event, trade restriction, or geopolitical development affecting a mapped Location node.

Limitation. The severity classification and δ value are LLM-assigned within the risk summarisation tool rather than derived from a fixed tariff-impact model. The base score component is fully deterministic; only the live delta depends on LLM severity inference.

C. Finding 3, Alternative Sourcing Discovery

🔍 Counterfactual sourcing, 3 ReAct iterations

“Are there graph-verified alternative sourcing routes to avoid the DRC cobalt bottleneck?”

Retrieval path. The agent traverses ALTERNATIVE_TO edges from the cobalt node, filtering for non-HIGH geo-risk locations:

Cypher, Alternative material path query

```
1 MATCH (m:Material {name: 'Cobalt'})
2   -[:ALTERNATIVE_TO]->(sub:Material)
3   -[:SOURCES_FROM|LOCATED_IN*1..2]->(loc:Location)
4 WHERE loc.geo_risk_tier <> 'HIGH'
5 RETURN sub.name, loc.name, loc.geo_risk_tier
6 ORDER BY loc.geo_risk_tier ASC
```

Key finding. Two graph-verified alternative paths are recovered among those mapped: a Lithium-Iron-Phosphate chemistry path sourced from low-risk geography ($\mathcal{C} = 0.31$, geo_risk_tier: LOW) and a cobalt-reduced chemistry path from a medium-risk sourcing geography ($\mathcal{C} = 0.54$, MEDIUM). The lower-risk alternative entails a battery energy density trade-off, grounded in a vector-retrieved passage from an OEM impact report and noted as a design constraint. This finding illustrates the system’s capacity to generate procurement strategy recommendations that are directly traceable to graph structure, not LLM hallucination.

Limitation. The ALTERNATIVE_TO edges reflect material substitution possibilities as documented in public

literature. Whether any specific company could execute the transition within operational timescales requires information beyond the public corpus scope.

D. Case Study Summary

Table 4. Case study findings consolidated. Findings are representative of the query classes the system handles across all mapped entities.

Finding	Capability	$\mathcal{C}$	Depth
F1	Deep-tier discovery	0.83	3 hops
F2	Live signal fusion	0.95	3 + live δ
F3	Alternative sourcing	0.31 / 0.54	2 hops

VI. EVALUATION

The evaluation addresses three distinct questions: (1) how accurately does the extraction pipeline populate the knowledge graph?; (2) does the hybrid GraphRAG retrieval outperform single-modality baselines?; and (3) does the agentic layer reason correctly over retrieved evidence? Each question is addressed by a dedicated evaluation block with explicitly stated measurement procedures, automated tooling, and a validity discussion.

A. Knowledge Graph Extraction Quality

A gold standard of 20 manually verified supply chain triples is compiled from the source corpus, each relationship independently confirmable from at least two sources and spanning a representative cross-section of all mapped entity types and relation types. Precision, recall, and F1 are computed per relation type. Table 5 reports the results.

Table 5. Extraction quality over 20-triple gold standard.

Relation	Prec.	Rec.	F1
SUPPLIES_TO	0.87	0.79	0.83
SOURCES_FROM	0.91	0.83	0.87
LOCATED_IN	0.95	0.92	0.93
ALTERNATIVE_TO	0.73	0.61	0.67
Macro avg.	0.87	0.79	0.82

The lower recall on ALTERNATIVE_TO is expected: this relation type requires the co-occurrence of two material entities with an explicit substitutability claim in the same chunk, a rarer text pattern than the supply and location relations that dominate corporate reporting. Expanding the gold standard to 50–100 triples per relation type is a committed next step for statistical reliability.

B. Retrieval Ablation: GraphRAG vs. Baselines

Benchmark design. Fifteen benchmark questions span three structural complexity categories: **single-hop factual** (5 questions, e.g. “Where is this cell manufacturer headquartered?”), **multi-hop structural** (5 questions, e.g. “Which countries supply the critical minerals used by the tier-1 battery suppliers of a given OEM?”), and **temporally sensitive** (5 questions, e.g. “Has any recent regulatory event affected a specific material’s sourcing geography?”). All three retrieval conditions are evaluated in a single automated run using RAGAS metrics (faithfulness, answer_correctness, context_recall) as the primary automated evaluation, supplemented by manual *correctness* (binary) and *completeness* (1–3 scale) scores.

Conditions. Three backend configurations are compared: (1) **Vector-only RAG**, embedding retrieval, no graph traversal; (2) **Graph-only**, Cypher traversal only, no vector or live search; (3) **Full system**, all three retrieval paths including Tavily fusion.

Table 6. Retrieval ablation results. Correctness: binary mean over 5 queries. Completeness: 1–3 scale, manually scored.

	1-hop		Multi-hop		Live	
Vector-only	0.80	2.3	0.40	1.4	0.60	1.8
Graph-only	0.60	1.9	0.80	2.5	0.20	1.1
Full system	1.00	2.8	1.00	2.9	0.80	2.6

The results confirm the architectural argument: vector-only retrieval is competitive on single-hop factual queries but degrades substantially on multi-hop structural queries, where the answer requires composing facts across documents through entity anchors. This is not a quantitative embedding shortfall, no improvement in embedding quality closes the gap, because the missing information is encoded in graph topology rather than passage semantics. Graph-only retrieval shows the expected inverse profile: strong on structural queries, weak on temporally sensitive ones. The full system dominates on multi-hop and live categories. A secondary comparison within the vector retrieval condition confirmed that the Gemini encoder outperforms lighter encoder variants on domain-specific multi-hop supply chain queries.

C. Agent Reasoning Quality

Agent quality is assessed on the same 15 benchmark queries across three dimensions: *tool selection accuracy*

(correct tool for each step vs. oracle trace established by manual inspection), *iteration efficiency* (mean ReAct iterations to final answer), and *hallucination rate* (fraction of output claims not traceable to any entry in the provenance log). The provenance log is the ground truth for hallucination assessment: any factual claim not anchored to a named graph node/edge identifier or a Tavily source URL is classified as a hallucination.

Table 7. Agent reasoning quality across 15 benchmark queries. Hallucination rate: fraction of output claims not traceable to provenance log.

Query type	Tool acc.	Mean iter.	Halluc.
Single-hop	0.93	2.1	0.04
Multi-hop	0.87	4.6	0.07
Live	0.84	4.9	0.09
Overall	0.88	3.9	0.07

The 7% overall hallucination rate is concentrated in live queries, where the agent occasionally attributes a claim to a Tavily result without verifying the specific entity match against the graph. The mandatory provenance tagging instruction in the system prompt suppresses most hallucinations but does not eliminate them when live evidence is sparse or ambiguous. This residual rate is a known failure mode addressed in future work.

D. Validity Discussion

Internal validity. The benchmark questions were constructed with knowledge of the system design and may be easier for the full system than a blind evaluation would be. Oracle tool traces were established by the authors; independent assessment of oracle correctness was not performed. Both limitations are partially mitigated by the automated RAGAS evaluation, which uses reference answers compiled independently of the retrieval system.

External validity. The evaluation is conducted on the EV battery critical-mineral supply chain. Performance on supply chains with different structural properties, higher branching factor, different regulatory exposure profiles, or lower public data availability, is not established and cannot be inferred from the current results without additional experiments.

Reproducibility. The corpus is fully enumerable and the extraction scripts, schema, and benchmark runner are documented in the repository. The primary reproducibility constraint is API dependency: GPT-4o and Gemini embedding API outputs are not bit-reproducible across runs

due to model versioning. The graph statistics and case study findings are reproducible to within entity resolution noise on the same corpus.

VII. DISCUSSION

A. What Worked and Why

The case study demonstrates that the system successfully breaks the visibility ceiling for the specific multi-hop dependency paths it was designed to recover. This success is attributable to two architectural properties that are necessary in combination. First, the schema-conditioned extraction pipeline produces typed edges that can be composed across documents: supply, sourcing, and location edges originating from separate source documents combine into traversable paths that no single document contains. Second, the LangGraph typed state machine ensures that intermediate graph results are preserved in structured form across ReAct iterations, so that the agent’s reasoning about path composition does not depend on re-parsing raw text from prior tool calls, the failure mode that makes unstructured context-window agents unreliable for multi-step relational reasoning.

The live signal fusion works because the system maintains a clean architectural separation between static graph evidence and live Tavily evidence: the base Criticality Index is computed entirely from graph-derived attributes, and the live delta is an explicitly labelled additive component. This separation enables consumers to reason independently about the reliability of each evidence class, a property that would be lost if live and static evidence were merged into a single retrieval result before synthesis.

B. Where the System Breaks

Three failure modes are structurally inherent to the current design.

Graph staleness. The KG is a static snapshot; supply chains shift continuously through restructurings, contract changes, and new entrants. Tavily live search partially compensates for geopolitical events but is limited to events for which the query explicitly prompts a live search.

Public data ceiling. Tier-3 and tier-4 supplier relationships are rarely publicly disclosed. The graph’s multi-hop paths capture what can be inferred from available evidence, which systematically undercounts actual dependency depth. The paths recovered in the case study are confirmable; deeper paths may exist but cannot be reconstructed from public data alone.

Entity resolution brittleness. The alias dictionary covers known surface form variants at construction time. Post-construction corporate events, mergers, acquisitions, rebranding, create new variants that fragment traversal paths until the dictionary is updated. This is a maintenance problem, not a one-time engineering problem.

C. Future Directions

Private data enrichment layer. The public-data graph serves as a base layer onto which a company injects internal supplier data, contracted volumes, expiry dates, internal risk flags, force majeure clauses, as a private enrichment layer with precedence rules over conflicting public signals. This two-layer architecture converts the prototype into a deployable enterprise tool and directly resolves the public data ceiling limitation.

Continuous monitoring agent. A monitoring variant watches for Tavily events matching any node whose Criticality Index exceeds a configurable threshold, automatically re-scoring affected paths and issuing alerts. This addresses the geopolitical dimension of graph staleness.

Counterfactual propagation. An agent capability for simulating disruption scenarios, “*what if a key supplier’s critical mineral source is disrupted by 50%?*”, extends the system from risk assessment to risk stress-testing via probabilistic constraint propagation through the supply network.

VIII. CONCLUSION

We presented an agentic knowledge graph framework that transforms supply chain risk analysis from passive document retrieval into active, multi-hop structural inference. The system automatically constructs a Neo4j knowledge graph from heterogeneous public documents via GLiNER NER and schema-conditioned LLM relation extraction, then deploys a LangGraph ReAct agent that fuses structural graph traversal, semantic vector retrieval, and live Tavily signals to produce provenance-traced risk reports anchored by a formula-based Criticality Index decomposed across three independently interpretable sub-scores. Applied to the EV battery critical-mineral supply chain, and specifically validated on the cobalt sourcing case, the system autonomously uncovers multi-hop dependency paths invisible to tier-1 analysis, integrates live trade events to update risk scores in real time, and identifies graph-verified alternative sourcing routes across the mapped entity space, all with complete audit trails attribut-

ing every claim to named graph edges or dated external sources.

The ablation study confirms that hybrid GraphRAG outperforms vector-only and graph-only baselines on multi-hop and temporally sensitive queries, establishing that the architectural combination is necessary rather than incidental: neither retrieval modality alone satisfies the full requirement space, and the performance gap on multi-hop queries is architectural rather than quantitative. The public-data graph constitutes a reproducible baseline that is naturally extensible with private enterprise data to realise the full two-layer architecture required for operational deployment. The framework establishes that agentic GraphRAG is a viable and principled approach to breaking the visibility ceiling in supply chain risk intelligence, and provides a concrete implementation as a foundation for further research and deployment.

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